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4. Vector Differentiation

Assignment-IV

1. Find the angle between the surfaces $x^2+y^2+z^2=9$ and $z=x^2+y^2-3$ at $(2,-1,2)$

Given surfaces

$$x^2+y^2+z^2=9 \quad , \quad x^2+y^2-3=z$$

$$x^2+y^2+z^2-9=0 \quad , \quad x^2+y^2-z-3=0$$

let point $P(2,-1,2)$

let $\phi_1 = x^2+y^2+z^2-9=0$ and $\phi_2 = x^2+y^2-z-3=0$

$\bar{n_1} = \nabla\phi_1$ at P , $\bar{n_2} = \nabla\phi_2$ at P

$$\nabla\phi_1 = \bar{i}\frac{\partial\phi_1}{\partial x} + \bar{j}\frac{\partial\phi_1}{\partial y} + \bar{k}\frac{\partial\phi_1}{\partial z}$$

$$= \bar{i}(2x) + \bar{j}(2y) + \bar{k}(2z)$$

$$(\nabla\phi_1)_P = 4\bar{i} - 2\bar{j} + 4\bar{k}$$

$$\therefore \bar{n_1} = 4\bar{i} - 2\bar{j} + 4\bar{k}$$

$$\nabla\phi_2 = \bar{i}\frac{\partial\phi_2}{\partial x} + \bar{j}\frac{\partial\phi_2}{\partial y} + \bar{k}\frac{\partial\phi_2}{\partial z}$$

$$= \bar{i}(2x) + \bar{j}(2y) + \bar{k}(-1)$$

$$(\nabla\phi_2)_P = 4\bar{i} - 2\bar{j} - \bar{k}$$

$$\therefore \bar{n_2} = 4\bar{i} - 2\bar{j} - \bar{k}$$

let θ be the angle between $\bar{n_1}$ and $\bar{n_2}$

$$\cos\theta = \frac{\bar{n_1}\cdot\bar{n_2}}{|\bar{n_1}||\bar{n_2}|} = \frac{16+4-4}{\sqrt{16+4+16}\sqrt{16+4+1}}$$

$$\cos\theta = \frac{16}{\sqrt{36}\sqrt{21}} = \frac{16}{6\sqrt{21}} = \frac{8}{3\sqrt{21}}$$

$$\theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)$$

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2. Find the angle between the surfaces $xy = z^2$ at $(4,1,2)$ and $(3,3,-3)$

Given surface

$$xy = z^2$$

let $\phi = xy - z^2$

let $P = (4,1,2)$, $Q = (3,3,-3)$

Normal vector $\bar{n_1}$ at P Normal vector $\bar{n_2}$ at Q

$\bar{n_1} = \nabla\phi$ at P , $\bar{n_2} = \nabla\phi$ at Q

$$\nabla\phi = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$$= \bar{i}(y) + \bar{j}(x) + \bar{k}(-2z)$$

$\bar{n_1} = \nabla\phi$ at $P = \bar{i} + 4\bar{j} - 4\bar{k}$, $\bar{n_2} = \nabla\phi$ at $Q = 3\bar{i} + 3\bar{j} + 6\bar{k}$

let θ be the angle between $\bar{n_1}$, $\bar{n_2}$

$$\cos\theta = \frac{\bar{n_1}\cdot\bar{n_2}}{|\bar{n_1}||\bar{n_2}|} = \frac{3+12-24}{\sqrt{1+16+16}\sqrt{9+9+36}} = \frac{-9}{\sqrt{33}\sqrt{54}}$$

$$\theta = \cos^{-1}\left(\frac{-9}{\sqrt{33}\sqrt{54}}\right)$$

3. Find the value of a and b such that the surfaces $ax^2 - byz = (a+2)x$ and $4x^2y + z^3 = 4$ cut orthogonally at $(1,-1,2)$

W.K.T

$$\cos\theta = \frac{\bar{n_1}\cdot\bar{n_2}}{|\bar{n_1}||\bar{n_2}|}$$

If normals are orthogonal $\theta = 90^\circ$

here $\cos 90^\circ = 0$

$$0 = \frac{\bar{n_1}\cdot\bar{n_2}}{|\bar{n_1}||\bar{n_2}|} \Rightarrow \bar{n_1}\cdot\bar{n_2} = 0$$

Given surfaces

$ax^2 - byz = (a+2)x$, $4x^2y + z^3 = 4$

$ax^2 - byz - (a+2)x = 0$ $4x^2y + z^3 - 4 = 0$

let $\phi_1 = ax^2 - byz - (a+2)x = 0$ $\phi_2 = 4x^2y + z^3 - 4 = 0$

$P = (1,1,-2)$

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$(4,1,2)$

$\bar{n_1} = \nabla\phi_1$ at P , $n\bar{2} = \nabla\phi_2$ at P

$\nabla\phi_1 = \bar{i}\frac{\partial\phi_1}{\partial x} + \bar{j}\frac{\partial\phi_1}{\partial y} + \bar{k}\frac{\partial\phi_1}{\partial z}$

$= \bar{i}(2ax - (a+2)) + \bar{j}(-bz) + \bar{k}(-by)$

$\bar{n_1} = \bar{i}(2a(1) - (a+2)) + \bar{j}(-b(-2)) + \bar{k}(-b(1))$

$= \bar{i}(a-2) + 2\bar{j}b - \bar{k}b$

$\nabla\phi_2 = \bar{i}\frac{\partial\phi_2}{\partial x} + \bar{j}\frac{\partial\phi_2}{\partial y} + \bar{k}\frac{\partial\phi_2}{\partial z}$

$= \bar{i}(8xy) + \bar{j}(4x^2) + \bar{k}(3z^2)$

$n\bar{2} = \bar{i}(8)(1)(1) + \bar{j}(4) + \bar{k}(3)(4)$

$= 8\bar{i} + 4\bar{j} + 12\bar{k}$

$\bar{n_1} \cdot \bar{n_2} = 0$

$\bar{n_1} \cdot n\bar{2} = 8(a-2) + 4(-2b) - 12(b) = 0$

$8a - 16 + 8b - 12b = 0$

$8a - 4b - 16 = 0$

$4a - 2b - 8 = 0$

$2a - b - 4 = 0$

clearly $P(1,1,-2)$ lies on $\phi_1 = c$ substitute b above

$a + 2b = a + 2$

$\boxed{b = 1}$

$2a - b = 4$

$2a - 1 = 4$

$2a = 5 \Rightarrow \boxed{a = 5/2}$

U Find the directional derivative of $f(x,y,z) = xy^3 + yz^3$ at $(2,-1,1)$ in the direction of the vector $\bar{i} + 2\bar{j} + 2\bar{k}$

Given

$f(x,y,z) = f = xy^3 + yz^3$

let $P = (2,-1,1)$ let $\bar{a} = \bar{i} + 2\bar{j} + 2\bar{k}$

$\nabla f = \bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}$

$= \bar{i}(y^3) + \bar{j}(3xy^2 + z^3) + \bar{k}(3z^2y)$

∇f at $P = \bar{i}(-1) + \bar{j}(3(2)(1) + 1) + \bar{k}(-3)$

$= -\bar{i} + \bar{j}(7) - 3\bar{k}$

let $\bar{e} = \hat{a} = \frac{\bar{a}}{|\bar{a}|} = \frac{\bar{i} + 2\bar{j} + 2\bar{k}}{\sqrt{1+4+4}} = \frac{\bar{i} + 2\bar{j} + 2\bar{k}}{3}$.

Required Directional derivation

$\bar{e} \cdot (\nabla f)$ at $P = \frac{\bar{i} + 2\bar{j} + 2\bar{k}}{3}(-\bar{i} + 7\bar{j} - 3\bar{k}) = \frac{-1 + 14 - 6}{3}$

$= 7/3$

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5 Find the directional derivative of $\phi(x,y,z) = xy^2 + yz^3$ at $(2,-1,1)$ in the direction of normal to the surface $x\log z - y^2 = -4$ at $(-1,2,1)$

Given $\phi(x,y,z) = \phi = xy^2 + yz^3$

let $P(2,-1,1)$

$$\nabla\phi = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$$= \bar{i}(y^2) + \bar{j}(2xy + z^3) + \bar{k}(3z^2y)$$

$$\nabla\phi \text{ at } P = \bar{i}(1) + \bar{j}(2(2)(-1)+1) + \bar{k}(3(1)(-1)$$

$$= \bar{i} + (-3\bar{j}) - 3\bar{k}$$

let Given surface be

$$f = x\log z - y^2 + 4 = 0$$

$$Q = (-1,2,1)$$

$$\bar{a} = \text{grad } f \text{ at } Q$$

$$\nabla f = \bar{i}(\log z) + \bar{j}(-2y) + \bar{k}\left(\frac{x}{z}\right)$$

$$\nabla f \text{ at } Q = \bar{i}(\log 1) + \bar{j}(-2(2)) + \bar{k}\left(\frac{-1}{1}\right)$$

$$= 0\bar{i} - 4\bar{j} - \bar{k}$$

Unit normal vector

$$\bar{e} = \frac{\bar{a}}{|\bar{a}|}$$

$$= \frac{-4\bar{j} - \bar{k}}{\sqrt{16+1}}$$

$$= \frac{-4\bar{j} - \bar{k}}{\sqrt{17}}$$

Required Directional Derivative

$$= \bar{e} \cdot (\nabla\phi) \text{ at } P$$

$$= \frac{1}{\sqrt{17}}(-4\bar{j} - \bar{k})(\bar{i} - 3\bar{j} - 3\bar{k})$$

$$= \frac{(12) + (3)}{\sqrt{17}}$$

$$= \frac{15}{\sqrt{17}}$$

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6 Find the directional derivative of $x^2-y^2+2z^2$ at the point $P(1,2,3)$ in the direction of the line PQ where Q is the point $(5,0,4)$ Also find the magnitude of the maximum directional derivative

Given $f = x^2-y^2+2z^2$

$P = (1,2,3)$

$\nabla f = \bar{i}(2x) + \bar{j}(-2y) + \bar{k}(4z)$

$= 2x\bar{i} - 2y\bar{j} + 4z\bar{k}$

(∇f) at $P = 2\bar{i} - 4\bar{j} + 12\bar{k}$

Given $P = (1,2,3)$ $\quad Q = (5,0,4)$

$\bar{OP} = \bar{i} + 2\bar{j} + 3\bar{k}$, $\quad \bar{OQ} = 5\bar{i} + 4\bar{k}$

$\bar{PQ} = \bar{OQ} - \bar{OP} = 4\bar{i} - 2\bar{j} + \bar{k}$

$|\bar{PQ}| = \sqrt{16+4+1} = \sqrt{21}$

$\bar{e} = \frac{\bar{PQ}}{|\bar{PQ}|} = \frac{4\bar{i}-2\bar{j}+\bar{k}}{\sqrt{21}}$

Required Directional Derivative

$= \bar{e} \cdot (\nabla\phi \text{ at } P)$

$= \frac{4\bar{i}-2\bar{j}+\bar{k}}{\sqrt{21}} (2\bar{i} - 4\bar{j} + 12\bar{k})$

$= \frac{8+8+12}{\sqrt{21}} = \frac{28}{\sqrt{21}}$

The magnitude of maximum directional derivative

$= |\nabla f \text{ at } P| = \sqrt{2^2+(-4)^2+12^2} = \sqrt{4+16+144} = \sqrt{164} = 2\sqrt{41}$

7 Find the directional derivative of $xy^2+yz^2+zx^2$ along the tangent to the curve $x=t, y=t^2, z=t^3$ at the point $(1,1,1)$

Given

let $\phi = xy^2+yz^2+zx^2$

let $P = (1,1,1)$

$\nabla\phi = \bar{i}(y^2+2xz) + \bar{j}(2xy+z^2) + \bar{k}(2zy+x^2)$

$\nabla\phi$ at $P = \bar{i}(1+2) + \bar{j}(2+1) + \bar{k}(2+1)$

$= 3\bar{i} + 3\bar{j} + 3\bar{k}$

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let $\bar{r}$ be the position vector at any point on the curve

$x=t,\ y=t^2,\ z=t^3$

$\bar{r} = x\bar{i}+y\bar{j}+z\bar{k}$

$\bar{r} = t\bar{i}+t^2\bar{j}+t^3\bar{k}$

Tangent formula $= \frac{d\bar{r}}{dt}$

$\frac{d\bar{r}}{dt} = \bar{i}+2t\bar{j}+3t^2\bar{k}$

$\left(\frac{d\bar{r}}{dt}\right)_{at\,P} = \left(\frac{d\bar{r}}{dt}\right)_{t=1} = \bar{i}+2\bar{j}+3\bar{k}$

$P(1,1,1)$

$x=t,\ y=t^2,\ z=t^3$

$t=1$	$t^2=1$	$t^3=1$
	$t=\pm 1$	$t=1, \omega, \omega^2$

$t=1$ satisfies all 3 eqns

let $\bar{e}$ be the unit vector then $\bar{e} = \frac{\bar{i}+2\bar{j}+3\bar{k}}{\sqrt{1+4+9}} = \frac{\bar{i}+2\bar{j}+3\bar{k}}{\sqrt{14}}$

Required $D\cdot D = \bar{e}\cdot(\nabla\phi \text{ at } P)$

$= \frac{(\bar{i}+2\bar{j}+3\bar{k})}{\sqrt{14}}(3\bar{i}+3\bar{j}+3\bar{k}) = \frac{3+6+9}{\sqrt{14}} = \frac{18}{\sqrt{14}}$

9. Find div $\bar{f}$ and curl $\bar{f}$, given that $\bar{f} = \text{grad}(x^3+y^3+z^3-3xyz)$

Given $\bar{f} = \text{grad}(x^3+y^3+z^3-3xyz)$

$= \bar{i}\frac{\partial}{\partial x}+\bar{j}\frac{\partial}{\partial y}+\bar{k}\frac{\partial}{\partial z}(x^3+y^3+z^3-3xyz)$

$= \bar{i}(3x^2-3yz)+\bar{j}(3y^2-3xz)+\bar{k}(3z^2-3xy)$

$\text{div}\,\bar{f} = \nabla\cdot\bar{f} = \bar{i}\frac{\partial}{\partial x}+\bar{j}\frac{\partial}{\partial y}+\bar{k}\frac{\partial}{\partial z}\left[\bar{i}(3x^2-3yz)+\bar{j}(3y^2-3xz)+\bar{k}(3z^2-3xy)\right]$

$= \frac{\partial}{\partial x}(3x^2-3yz)+\frac{\partial}{\partial y}(3y^2-3xz)+\frac{\partial}{\partial z}(3z^2-3xy)$

$= 6x+6y+6z \Rightarrow 6(x+y+z)$

$\text{curl}\,\bar{f} = \nabla\times\bar{f} \Rightarrow \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3x^2-3yz & 3y^2-3xz & 3z^2-3xy \end{vmatrix}$

$= \bar{i}(-3x+3x)-\bar{j}(-3y+3y)+\bar{k}(-3z+3z)$

$= \bar{i}(0)-\bar{j}(0)+\bar{k}(0)$

$= \bar{0}$

$\therefore$ div $\bar{f} = 6(x+y+z)$ and curl $\bar{f} = \bar{0}$

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10. Find div$\bar{F}$ and curl$\bar{F}$ at $(1,-1,1)$ given that $\bar{F} = xy^2\bar{i} + 2x^2yz\bar{j} - 3yz^2\bar{k}$

Given $\bar{f} = xy^2\bar{i} + 2x^2yz\bar{j} - 3yz^2\bar{k}$

let $P = (1,-1,1)$

$$\text{div}\,\bar{f} = \nabla \cdot \bar{f}$$

$$= \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)\cdot\left(xy^2\bar{i} + 2x^2yz\bar{j} - 3yz^2\bar{k}\right)$$

$$= \frac{\partial}{\partial x}xy^2 + \frac{\partial}{\partial y}2x^2yz + \frac{\partial}{\partial z}(-3yz^2)$$

$$= y^2 + 2x^2z - 6yz$$

$$(\text{div}\,\bar{f})_{at\,P} = 1 + 2 + 6(-1) \Rightarrow 1 + 2 + 6 = 9$$

$$\text{curl}\,\bar{f} = \nabla \times \bar{f}$$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy^2 & 2x^2yz & -3yz^2 \end{vmatrix} \Rightarrow$$

$$= \bar{i}(-3z^2 - 2x^2y) - \bar{j}(0-0) + \bar{k}(4xyz - 2xy)$$

$$(\text{curl}\,\bar{f})_{at\,P} = \bar{i}(-3+2) - \bar{j}(0) + \bar{k}(-4+2)$$

$$= -\bar{i} + 0\bar{j} - 2\bar{k}$$

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11. Prove that $\text{div}(r^n\bar{r}) = (n+3)r^n$, hence show that $\frac{\bar{r}}{r^3}$ is solenoidal.

Given let $\bar{f} = r^n\bar{r}$

$$= r^n(x\bar{i} + y\bar{j} + z\bar{k})$$

$$= xr^n\bar{i} + yr^n\bar{j} + zr^n\bar{k}$$

consider $\text{div}(\bar{f}) = \text{div}(r^n\bar{r})$

$$= \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)\cdot\left(xr^n\bar{i} + yr^n\bar{j} + zr^n\bar{k}\right)$$

$$= \frac{\partial}{\partial x}(xr^n) + \frac{\partial}{\partial y}(yr^n) + \frac{\partial}{\partial z}(zr^n)$$

$$= \sum \frac{\partial}{\partial x}(xr^n)$$

$$= \sum xnr^{n-1}\left(\frac{\partial r}{\partial x}\right) + r^n(1)$$

$= \sum x n r^{n-1}\left(\frac{x}{r}\right) + r^n$

$= \sum n x^2 r^{n-2} + r^n$

$= n x^2 r^{n-2} + r^n + n y^2 r^{n-2} + r^n + n z^2 r^{n-2} + r^n$

$= 3r^n + n r^{n-2}(x^2+y^2+z^2)$

$= 3r^n + n r^{n-2}(r^2)$

$= 3r^n + n r^n \Rightarrow (n+3) r^n$

$\therefore \underset{\text{div } r^n \bar{r}}{\text{div}\, \bar{f}} = (n+3) r^n$ — (1)

Consider $\text{div}\left(\frac{\bar{r}}{r^3}\right) \Rightarrow \text{div}(r^{-3}\bar{r})$

from (1)

$\text{div}(r^{-3}\bar{r}) = (-3+3) r^{-3}$

$= 0$

$\therefore \frac{\bar{r}}{r^3}$ is solenoidal

12. Show that the vector field $\bar{F} = 2xyz^2\bar{i} + (x^2z^2 + z\cos yz)\bar{j} + (2x^2yz + y\cos yz)\bar{k}$ is irrotational and find the potential function.

Given $\bar{F} = 2xyz^2\bar{i} + (x^2z^2 + z\cos yz)\bar{j} + (2x^2yz + y\cos yz)\bar{k}$

consider

$\text{curl}\, \bar{f} = \nabla \times \bar{f}$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2xyz^2 & x^2z^2 + z\cos yz & 2x^2yz + y\cos yz \end{vmatrix}$$

$= \bar{i}\left[(2x^2 + y(-\sin yz)z + \cos yz(1) - 2x^2z - z(-\sin yz)y - \cos yz)\right] - \bar{j}(4xyz - 4xyz) + \bar{k}(2xz^2 - 2xz^2)$

$= \bar{i}(0) - \bar{j}(0) + \bar{k}(0)$

$\text{curl}\, \bar{f} = \bar{0}$

$\therefore \bar{f}$ is irrotational

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If $\bar{F}$ is irrotational then there exists a scalar function ϕ such that $\bar{F} = \nabla\phi$

$$2xyz^2\,\bar{i} + (x^2z^2 + z\cos yz)\,\bar{j} + (2x^2yz + y\cos yz)\,\bar{k} = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$\frac{\partial\phi}{\partial x} = 2xyz^2$

Iobs wrt x

$\phi = \frac{2x^2}{2}yz^2 + c_1$

$\frac{\partial\phi}{\partial y} = x^2z^2 + z\cos yz$

$\bar{j}$ obs wrt y

$\phi = x^2yz^2 + \frac{z\sin yz}{z} + c_2$

$\frac{\partial\phi}{\partial z} = 2x^2yz + y\cos yz$

Iobs wrt z

$\phi = \frac{2x^2yz^2}{2} + \frac{y\sin yz}{y} + c_3$

Scalar potential $\therefore \phi = x^2yz^2 + \sin yz + c$

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13. Show that the vector field $\bar{F} = (x^2 - yz)\bar{i} + (y^2 - xz)\bar{j} + (z^2 - xy)\bar{k}$ is irrotational and hence find the scalar potential function corresponding to it.

Given

$$\bar{F} = (x^2 - yz)\bar{i} + (y^2 - xz)\bar{j} + (z^2 - xy)\bar{k}$$

consider $\text{curl}\,\bar{f} = \nabla \times \bar{F} =$

$$\begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 - yz & y^2 - xz & z^2 - xy \end{vmatrix}$$

$\text{curl}\,\bar{f} = \bar{i}(-x + x) - \bar{j}(-y + y) + \bar{k}(-z + z)$

$= \bar{0}$

$\therefore \bar{F}$ is irrotational

If $\bar{F}$ is an irrotational vector then there exist a scalar potential function ϕ such that $\bar{F} = \nabla\phi$

$$\bar{F} = \nabla\phi$$

$$(x^2 - yz)\bar{i} + (y^2 - xz)\bar{j} + (z^2 - xy)\bar{k} = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$\frac{\partial\phi}{\partial x} = x^2 - yz$

Iobs wrt x

$\phi = x^3/3 - yzx + c_1$

$\frac{\partial\phi}{\partial y} = y^2 - xz$

Iobs wrt y

$\phi = y^3/3 - xyz + c_2$

$\frac{\partial\phi}{\partial z} = z^2 - xy$

Iobs wrt z

$\phi = z^3/3 - xyz + c_3$

$\therefore \phi = x^3/3 + y^3/3 + z^3/3 - xyz + c.$

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14. Find the constants a, b, c so that the following vector $\bar{A}$ is irrotational. Also find its scalar potential function. Given $\bar{A} = (x+2y+az)\bar{i} + (bx-3y-z)\bar{j} + (4x+cy+2z)\bar{k}$

Given $\bar{A} = [x+2y+az]\bar{i} + [bx-3y-z]\bar{j} + [4x+cy+2z]\bar{k}$

$\text{curl } \bar{A} = \nabla \times \bar{A}$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x+2y+az & bx-3y-z & 4x+cy+2z \end{vmatrix}$$

$= \bar{i}(c+1) - \bar{j}(4-a) + \bar{k}(b-2)$

$= \bar{i}(c+1) - \bar{j}(4-a) + \bar{k}(b-2)$

$\bar{A}$ is irrotational so $\nabla \times \bar{A} = \bar{0}$

$\bar{i}(c+1) - \bar{j}(4-a) + \bar{k}(b-2) = \bar{0}$ $\quad \{\because \bar{0} = 0\bar{i} + 0\bar{j} + 0\bar{k}$

$c+1=0 \Rightarrow c=-1$ | $4-a=0 \Rightarrow a=4$ | $b-2=0 \Rightarrow b=2$

If $\bar{A}$ is an irrotational vector then there exists a scalar function ϕ such that $\bar{A} = \nabla\phi$

$\bar{A} = (x+2y+4z)\bar{i} + (2x-3y-z)\bar{j} + (4x+cy+2z)\bar{k} = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$

$\frac{\partial\phi}{\partial x} = x+2y+4z$ | $\frac{\partial\phi}{\partial y} = 2x-3y-z$ | $\frac{\partial\phi}{\partial z} = 4x-y+2z$

IOBS w.r.t x | IOBS w.r.t x | IOBS w.r.t z

$\phi = \frac{x^2}{2} + 2xy + 4xz + C_1$ | $\phi = 2xy - \frac{3y^2}{2} - zy + C_2$ | $\phi = 4xz - yz + \frac{2z^2}{2} + C_3$

$\phi = x^2/2 - 3y^2/2 + z^2 + 2xy + 4xz - yz + C$

15. Prove that (i) $\text{div}(\text{curl}\,\bar{f}) = 0$ (ii) $\text{curl}(\text{grad}\,\phi) = \bar{0}$

i) $\text{div}(\text{curl}\,\bar{f}) = 0$

consider $\text{curl}\,\bar{f} = \nabla \times \bar{f}$

let $\bar{f} = f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$

$$\text{curl}\,\bar{f} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f_1 & f_2 & f_3 \end{vmatrix}$$

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$= \bar{i}\left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right) - \bar{j}\left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z}\right) + \bar{k}\left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)$

$\text{div}(\text{curl}\,\bar{f}) = \nabla \cdot \text{curl}\,\bar{f}$

$= \frac{\partial}{\partial x}\left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right) - \frac{\partial}{\partial y}\left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z}\right) + \frac{\partial}{\partial z}\left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)$

$= \frac{\partial^2 f_3}{\partial x \partial y} - \frac{\partial^2 f_2}{\partial x \partial z} - \frac{\partial^2 f_3}{\partial y \partial x} + \frac{\partial^2 f_1}{\partial y \partial z} + \frac{\partial^2 f_2}{\partial z \partial x} - \frac{\partial^2 f_1}{\partial z \partial y} = 0.$

ii) $\text{curl}\,\text{grad}\,\phi = \bar{0}$

consider $\text{grad}\,\phi = \nabla\phi$

$= \bar{i}\frac{\partial \phi}{\partial x} + \bar{j}\frac{\partial \phi}{\partial y} + \bar{k}\frac{\partial \phi}{\partial z}$

$\text{curl}(\text{grad}\,\phi) = \nabla \times \text{grad}\,\phi$

$= \nabla \times \nabla\phi$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial\phi/\partial x & \partial\phi/\partial y & \partial\phi/\partial z \end{vmatrix}$$

$= \bar{i}\left[\frac{\partial^2\phi}{\partial z \partial y} - \frac{\partial^2\phi}{\partial y \partial z}\right] - \bar{j}\left[\frac{\partial^2\phi}{\partial z \partial x} - \frac{\partial^2\phi}{\partial x \partial z}\right] + \bar{k}\left[\frac{\partial^2\phi}{\partial x \partial y} - \frac{\partial^2\phi}{\partial y \partial x}\right]$

$= \bar{i}(0) - \bar{j}(0) + \bar{k}(0)$

$= \bar{0}$

16 show that $\text{div}(\text{grad}\,r^n) = \nabla^2(r^n) = n(n+1)\,r^{n-2}$ where $r = |\bar{r}|$

let $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

$r = |\bar{r}| = \sqrt{x^2+y^2+z^2}$, $r^2 = x^2 + y^2 + z^2$

$\frac{\partial r}{\partial x} = \frac{x}{r}$, $\frac{\partial r}{\partial y} = \frac{y}{r}$, $\frac{\partial r}{\partial z} = \frac{z}{r}$

$\text{div}(\text{grad}\,r^n) = \nabla \cdot \text{grad}(r^n)$

$= \nabla \cdot \nabla(r^n)$ — ①

$= \nabla^2(r^n)$

consider $\nabla r^n = \bar{i}\frac{\partial}{\partial x}(r^n) + \bar{j}\frac{\partial}{\partial y}(r^n) + \bar{k}\frac{\partial}{\partial z}(r^n)$

$= \sum \bar{i}\frac{\partial}{\partial x}(r^n)$

$= \sum \bar{i}\, n r^{n-1}\frac{\partial r}{\partial x}$ $\Rightarrow \sum \bar{i}\, n r^{n-1}\left(\frac{x}{r}\right)$

$= \sum \bar{i}\, nx r^{n-2}$

$\nabla r^n = nx r^{n-2}\bar{i} + ny r^{n-2}\bar{j} + nz r^{n-2}\bar{k}$

from ①

$\nabla\cdot(\nabla r^n) = \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)\cdot\left(nx r^{n-2}\bar{i} + ny r^{n-2}\bar{j} + nz r^{n-2}\bar{k}\right)$

$= \sum \frac{\partial}{\partial x}\, nx r^{n-2}$

$= \sum n\left[x(n-2) r^{n-3}\left(\frac{\partial r}{\partial x}\right) + r^{n-2}\right]$

$= \sum n\left[(n-2)\, x\, r^{n-3}\left(\frac{x}{r}\right) + r^{n-2}\right]$

$= \sum n\left[x^2(n-2) r^{n-4} + r^{n-2}\right]$

$= n\left[x^2(n-2) r^{n-4} + r^{n-2}\right] + n\left[y^2(n-2) r^{n-4} + r^{n-2}\right] + n\left[z^2(n-2) r^{n-4} + r^{n-2}\right]$

$= n(n-2) r^{n-4}\left[x^2+y^2+z^2\right] + 3n r^{n-2}$

$= n(n-2) r^{n-4}(r^2) + 3n r^{n-2}$

$= n(n-2) r^{n-2} + 3n r^{n-2}$

$= r^{n-2}\left[n(n-2) + 3n\right]$

$= n r^{n-2}\left[n-2+3\right]$

$= n(n+1) r^{n-2}$

$\therefore \operatorname{div}(\operatorname{grad} r^n) = \nabla^2 r^n = n(n+1) r^{n-2}$

17. Show that $\nabla^2[f(r)] = f''(r) + \frac{2}{r} f'(r)$ where $r = |\bar{r}|$

let $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

$r^2 = x^2 + y^2 + z^2$

$\frac{\partial r}{\partial x} = \frac{x}{r}$, $\frac{\partial r}{\partial y} = \frac{y}{r}$, $\frac{\partial r}{\partial z} = \frac{z}{r}$

$\nabla^2[f(r)] = \nabla\cdot[\nabla f(r)]$ — ①

consider $\nabla f(r) = \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right) f(r)$

$= \sum \bar{i}\, \frac{\partial f(r)}{\partial x}$

$= \sum \bar{i}\, f'(r)\frac{\partial r}{\partial x}$

$= \sum \bar{i}\, f'(r)\frac{x}{r}$

$= \frac{x}{r} f'(r)\bar{i} + \frac{y}{r} f'(r)\bar{j} + \frac{z}{r} f'(r)\bar{k}$

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$$= \sum \bar{i} f'(r)\frac{x}{r}$$

$$= \bar{i} f'(r)\frac{x}{r} + \bar{j} f'(r)\frac{y}{r} + \bar{k} f'(r)\frac{z}{r}$$

from ①

$$\nabla^2 f(r) = \nabla [\nabla f(r)]$$

$$= \left[\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right]\left[\bar{i} f'(r)\frac{x}{r} + \bar{j} f'(r)\frac{y}{r} + \bar{k} f'(r)\frac{z}{r}\right]$$

$$= \frac{\partial}{\partial x} f'(r)\frac{x}{r} + \frac{\partial}{\partial y} f'(r)\frac{y}{r} + \frac{\partial}{\partial z} f'(r)\frac{z}{r}$$

$$= \sum \frac{\partial}{\partial x} f'(r)\frac{x}{r}$$

$$= \sum \frac{r\left(f'(r)(1) + x f''(r)\frac{\partial r}{\partial x}\right) - f'(r)(x)\cdot\frac{\partial r}{\partial x}}{r^2}$$

$$= \sum \frac{1}{r^2}\left[r f'(r) + x^2 f''(r)\frac{x}{r} - f'(r)\, x\left(\frac{x}{r}\right)\right]$$

$$= \sum \frac{1}{r^2}\left[r f'(r) + f''(r)\, x^2 - \frac{f'(r)}{r} x^2\right]$$

$$= \frac{1}{r^2}\left[r f'(r) + f''(r)x^2 - \frac{f'(r)}{r}x^2 + r f'(r) + f''(r) y^2 - \frac{f'(r)}{r} y^2 + r f'(r) + f''(r) z^2 - \frac{f'(r)}{r} z^2\right]$$

$$= \frac{1}{r^2}\left[3r f'(r) + f''(r)\left[x^2+y^2+z^2\right] - \frac{f'(r)}{r}\left[x^2+y^2+z^2\right]\right]$$

$$= \frac{1}{r^2}\left[3r f'(r) + f''(r)(r^2) - f'(r)\, r^2\right]$$

$$= \frac{1}{r^2}\left[2r f'(r) + f''(r)(r^2)\right]$$

$$= \frac{2r f'(r)}{r^2} + f''(r) \Rightarrow \frac{2}{r} f'(r) + f''(r)$$

$$\therefore \nabla^2 f(r) = \frac{2}{r} f'(r) + f''(r)$$

8. The temperature of points in a space is given by $T(x,y,z) = x^2+y^2-z$, a mosquito is located at $(1,1,2)$ desires to fly in such a direction that it will get warm as soon as possible. In what direction should it move. Also find what is the maximum rate of change in the temperature

Given

$T(x,y,z) = x^2+y^2-z$.

mosquito's position, let $P=(1,1,2)$

(The direction in which a scalar function $T(x,y,z)$ increases most rapidly is gradient vector i.e $(\nabla T)_{atp}$)

$$\nabla T = \frac{\partial T}{\partial x}\bar{i} + \frac{\partial T}{\partial y}\bar{j} + \frac{\partial T}{\partial z}\bar{k}$$

$$= \frac{\partial}{\partial x}(x^2+y^2-z)\bar{i} + \frac{\partial}{\partial y}(x^2+y^2-z)\bar{j} + \frac{\partial}{\partial z}(x^2+y^2-z)\bar{k}$$

$$= 2x\bar{i} + 2y\bar{j} - \bar{k}$$

$$\nabla T_{atp} = 2(1)\bar{i} + 2(1)\bar{j} - \bar{k}$$

$$= 2\bar{i} + 2\bar{j} - \bar{k}$$

∴ The mosquito should move in the direction $2\bar{i}+2\bar{j}-\bar{k}$ (In unit vector form $v = 2/3\,\bar{i} - 2/3\,\bar{j} - \frac{1}{3}\bar{k}$)

(The max rate of change in temp is equal to the magnitude of the gradient vector, $|\nabla T|$)

$$|\nabla T_{atp}| = \sqrt{2^2+2^2+(-1)^2}$$

$$= \sqrt{4+4+1}$$

$$= \sqrt{9}$$

$$= 3$$

∴ The maximum rate of change is 3 units per unit distance.

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